

Apache Hive

Practical Assignment

Subject: Big Data Analytics | Database: A2 | Tables: flight, flight1

Section 1 — Create Database & Table

Step 1: Launch Hive Shell

Open the terminal and start the Hive interactive shell.

```
hive
```

Step 2: Create and Use Database

Create a new database named A2 and switch into it.

```
create database A2;
use A2;
```

Step 3: Create Table (Basic)

Define the flight table schema and verify its structure with DESC.

```
create table flight(fno int, year int, des varchar(20), delay float);

desc flight;

-- Drop the basic table before recreating with proper format
drop table flight;
```

Step 4: Recreate Table with Proper Format

Recreate the table specifying comma-separated format and text file storage — required for LOAD DATA operations.

```
create table flight(fno int, year int, des varchar(20), delay float)
row format delimited
fields terminated by ','
lines terminated by '\n'
stored as textfile;
```

Section 2 — Insert Data & Query Table: flight

Step 5: Insert Records

Insert three flight records directly using INSERT INTO.

```
insert into flight values(101, 2005, "Mumbai", 30.6);
insert into flight values(102, 2006, "Pune", 10.2);
```

```
insert into flight values(103, 2007, "Kolkatta", 9.6);
```

Step 6: View All Records

Select all records from the flight table.

```
use A2;
select * from flight;
```

Expected Output — flight table

fno	year	des	delay
101	2005	Mumbai	30.6
102	2006	Pune	10.2
103	2007	Kolkatta	9.6

Section 3 — Create External File & Load into flight1

Step 7: Create the External Text File

Open a text editor (gedit) and save the file as f.txt on the Desktop. The file contains source city data for each flight.

```
# Open text editor
gedit f.txt

# Check working directory and file listing
pwd
ls
```

Contents of f.txt

```
# File: /home/cloudera/Desktop/f.txt
101,2005,Nashik
102,2006,Nagpur
103,2007,Treni
```

Step 8: Create Table flight1

Create the flight1 table that holds flight number, year, and source city.

```
use A2;

create table flight1(fno int, year int, src varchar(20))
row format delimited
fields terminated by ','
lines terminated by '\n'
stored as textfile;

desc flight1;
```

Step 9: Load Data from File into flight1

Use LOAD DATA LOCAL INPATH to import f.txt into the flight1 table.

```
LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/f.txt'
OVERWRITE INTO TABLE flight1;
```

Step 10: Verify Loaded Data

```
select * from flight1;
```

Expected Output — flight1 table

fno	year	src
101	2005	Nashik
102	2006	Nagpur
103	2007	Treni

Section 4 — JOIN Query (flight1 ■ flight)

Step 11: Join Both Tables

Perform an INNER JOIN on flight and flight1 using fno and year as keys to get a combined view of source city, destination, and delay.

```
SELECT f1.fno, f1.year, f1.src, f.des, f.delay
FROM flight1 f1
JOIN flight f
ON f1.fno = f.fno AND f1.year = f.year;
```

Expected Output — JOIN Result

fno	year	src	des	delay
101	2005	Nashik	Mumbai	30.6
102	2006	Nagpur	Pune	10.2
103	2007	Treni	Kolkatta	9.6

Quick Reference — All Commands

```
-- SETUP
hive
create database A2;
use A2;

-- TABLE: flight
create table flight(fno int, year int, des varchar(20), delay float)
row format delimited fields terminated by ',' lines terminated by '\n'
```

```
stored as textfile;
insert into flight values(101,2005,"Mumbai",30.6);
insert into flight values(102,2006,"Pune",10.2);
insert into flight values(103,2007,"Kolkatta",9.6);
select * from flight;

-- TABLE: flight1 (from file)
create table flight1(fno int, year int, src varchar(20))
row format delimited fields terminated by ',' lines terminated by '\n'
stored as textfile;
LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/f.txt' OVERWRITE INTO TABLE
flight1;
select * from flight1;

-- JOIN
SELECT f1.fno, f1.year, f1.src, f.des, f.delay
FROM flight1 f1 JOIN flight f
ON f1.fno = f.fno AND f1.year = f.year;
```

Note: Ensure Hadoop and Hive services are running before executing these commands. The file path /home/cloudera/Desktop/f.txt assumes a Cloudera QuickStart VM environment.